OisteBio GmbH
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Conversion Log

Production Conversion Log

July 2025 · Girardot, Colombia

Submission reference	`RTFO-310825`
Reporting period	July 2025 (2025-07)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	`0,774 kg/L`
Density — PLUS-grade	`0,871 kg/L`
Production days observed	27 of 31 (4 non-production days)
Source	`daily_production` table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

____
Paolo Ughetti
Managing Director

Date: ____________________

Executive Summary

Per-day kilogram-to-litre conversion log for **July 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	3 778 672,000 kg
Total EU-grade production	1 144 937,616 kg / 1 479 247,564 L
Total PLUS-grade production	1 510 007,716 kg / 1 733 648,354 L
Liquid-product yield (EU + PLUS) over input	70,26 %
Grand total volume (EU + PLUS)	**3 212 895,918 L**

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **516 610,677 kg**, steel scrap **317 049,463 kg**, water condensate **57 338,733 kg**, syngas **160 211,170 kg**, recorded losses **72 516,625 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — July 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,774 kg/L**, **PLUS 0,871 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
`2025-07-01`	142 164,000	43 075,692	56 810,630	`0,774`	`0,871`	55 653,349	65 224,604	70,26 %	—
`2025-07-02`	149 075,000	45 169,725	59 572,358	`0,774`	`0,871`	58 358,818	68 395,359	70,26 %	—
`2025-07-03`	131 827,000	39 943,581	52 679,827	`0,774`	`0,871`	51 606,694	60 482,006	70,26 %	—
`2025-07-04`	142 728,000	43 246,584	57 036,012	`0,774`	`0,871`	55 874,140	65 483,366	70,26 %	—
Total	**3 778 672,000**	**1 144 937,616**	**1 510 007,716**	—	—	**1 479 247,564**	**1 733 648,354**	**70,26 %**	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-07-05	127 754,000	38 709,462	51 052,202	0,774	0,871	50 012,225	58 613,320	70,26 %	—
2025-07-07	141 110,000	42 756,330	56 389,437	0,774	0,871	55 240,736	64 741,030	70,26 %	—
2025-07-08	149 332,000	45 247,596	59 675,058	0,774	0,871	58 459,426	68 513,270	70,26 %	—
2025-07-09	153 499,000	46 510,197	61 340,247	0,774	0,871	60 090,694	70 425,083	70,26 %	—
2025-07-10	137 432,000	41 641,896	54 919,660	0,774	0,871	53 800,899	63 053,571	70,26 %	—
2025-07-11	145 415,000	44 060,745	58 109,773	0,774	0,871	56 926,027	66 716,157	70,26 %	—
2025-07-12	144 214,000	43 696,842	57 629,837	0,774	0,871	56 455,868	66 165,140	70,26 %	—
2025-07-14	152 311,000	46 150,233	60 865,506	0,774	0,871	59 625,624	69 880,030	70,26 %	—
2025-07-15	148 954,000	45 133,062	59 524,004	0,774	0,871	58 311,450	68 339,844	70,26 %	—
2025-07-16	138 838,000	42 067,914	55 481,516	0,774	0,871	54 351,310	63 698,641	70,26 %	—
2025-07-17	137 491,000	41 659,773	54 943,237	0,774	0,871	53 823,996	63 080,639	70,26 %	—
2025-07-18	149 873,000	45 411,519	59 891,250	0,774	0,871	58 671,213	68 761,481	70,26 %	—
2025-07-19	123 856,000	37 528,368	49 494,509	0,774	0,871	48 486,264	56 824,924	70,26 %	—
2025-07-21	137 174,000	41 563,722	54 816,559	0,774	0,871	53 699,899	62 935,200	70,26 %	—
2025-07-22	136 960,000	41 498,880	54 731,042	0,774	0,871	53 616,124	62 837,017	70,26 %	—
2025-07-23	141 351,000	42 829,353	56 485,745	0,774	0,871	55 335,081	64 851,602	70,26 %	—
2025-07-24	131 788,000	39 931,764	52 664,242	0,774	0,871	51 591,426	60 464,113	70,26 %	—
2025-07-25	131 003,000	39 693,909	52 350,546	0,774	0,871	51 284,120	60 103,956	70,26 %	—
2025-07-26	111 795,000	33 873,885	44 674,773	0,774	0,871	43 764,709	51 291,358	70,26 %	—
2025-07-28	149 239,000	45 219,417	59 637,895	0,774	0,871	58 423,019	68 470,603	70,26 %	—
2025-07-29	137 199,000	41 571,297	54 826,550	0,774	0,871	53 709,686	62 946,670	70,26 %	—
2025-07-30	143 850,000	43 586,550	57 484,378	0,774	0,871	56 313,372	65 998,138	70,26 %	—
Total	**3 778 672,000**	**1 144 937,616**	**1 510 007,716**	—	—	**1 479 247,564**	**1 733 648,354**	**70,26 %**	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-07-31	142 440,000	43 159,320	56 920,923	0,774	0,871	55 761,395	65 351,232	70,26 %	—
Total	**3 778 672,000**	**1 144 937,616**	**1 510 007,716**	**—**	**—**	**1 479 247,564**	**1 733 648,354**	**70,26 %**	

Non-production days (4): 2025-07-06, 2025-07-13, 2025-07-20, 2025-07-27. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during July 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform `product_densities` table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to July 2025 are:

- **EU-grade (DEV-P100-EU)** — ρ_{EU} = 0,774 kg/L
- **PLUS-grade (DEV-P100-PLUS)** — ρ_{PLUS} = 0,871 kg/L

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform `product_densities` table — initially loaded by Alembic migration `0005_seed_product_densities.py` and extended with per-month effective rows by `0014_monthly_densities.py`. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu   NUMERIC GENERATED ALWAYS AS (eu_prod_kg   / 0.774)  STORED,
litres_plus NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.871) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code `42601`.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix `000000000000`). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

— End of Production Conversion Log —